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"""AeroPulse-X – Adaptive Airport Resilience Intelligence (academic research prototype).

Run: streamlit run app.py
"""
import sys
from pathlib import Path

ROOT = Path(__file__).resolve().parent
sys.path.insert(0, str(ROOT))

import streamlit as st # noqa: E402

st.set_page_config(page_title="AeroPulse-X", page_icon="✈", layout="wide",
initial_sidebar_state="expanded")

from src import ui # noqa: E402
from src.system import AeroSystem # noqa: E402


@st.cache_resource(show_spinner="First launch: training AeroPulse-X on simulated airport data (about
1–2 minutes)…")
def load_system():
s = AeroSystem.load()
if s is None:
s = AeroSystem.build(evaluate=False, save=True, progress=lambda m: None)
return s


st.markdown(ui.CSS, unsafe_allow_html=True)
system = load_system()
st.session_state["ctx"] = ui.sidebar_and_ctx(system)
st.session_state["system"] = system

PAGES = [
("pages/01_command_center.py", "Command Center", "■"),
("pages/02_operational_dna.py", "Airport Operational DNA", "■"),
("pages/03_airport_stress.py", "Airport Stress", "■"),
("pages/04_interaction_graph.py", "Interaction Graph", "■■"),
("pages/05_early_warning.py", "Early Warning", "■"),
("pages/06_future_airport.py", "Future Airport", "■"),
("pages/07_disruption_ripple.py", "Disruption Ripple", "■"),
("pages/08_whatif_lab.py", "What-If Laboratory", "■"),
("pages/09_counterfactual.py", "Counterfactual Analysis", "■"),
("pages/10_intervention_optimizer.py", "AI Intervention Optimizer", "■"),
("pages/11_resilience.py", "Resilience Analysis", "■■"),
("pages/12_explainable_ai.py", "Explainable AI", "■"),
("pages/13_stress_testing.py", "Stress Testing", "■"),
("pages/14_model_performance.py", "Model Performance", "■"),
("pages/15_simulation_history.py", "Simulation History", "■"),
("pages/16_research_summary.py", "Research Summary", "■"),
]
```

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nav = st.navigation([st.Page(p, title=t, icon=i, default=(n == 0)) for n, (p, t, i) in enumerate(PAGES)])
nav.run()
```